

Mathews Shaji

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PROFESSIONAL SUMMARY

Backend-leaning full-stack developer with production-grade SaaS and cloud-native experience from the ground up. Specialised in **Python/FastAPI backends**, **distributed services**, **multi-tenant architecture**, and **AWS/GCP/Azure integrations**, complemented by modern **Next.js/React frontends**. Has independently shipped complex platform features including cloud onboarding workflows, data ingestion pipelines, operational dashboards, authentication systems, and containerised deployments in fast-moving startup environments. Proven track record building **AI/LLM integrations** (OpenAI, Amazon Bedrock, LangChain) with production-grade pipelines including structured output validation and per-tenant cost attribution.

EXPERIENCE

Full-Stack Developer

Thoughtroutes

Nov 2023 – Present
Remote

CloudPositive – Multi-Cloud Cost Management Platform

Nov 2023 – Present

- Architected multi-tenant **FastAPI** backend with dynamic per-tenant MySQL schema isolation and role-specific dashboards for finance, BU leader, and executive personas
- Designed and built a data ingestion engine with 20+ modular pipelines covering **AWS** (Cost Explorer, Trusted Advisor, Compute Optimiser), **GCP** (BigQuery billing export), and **Azure** Cost Management, orchestrated via AWS SQS
- Engineered secretless multi-cloud onboarding across all three clouds: AWS (IAM roles, no stored credentials), GCP (service account BigQuery billing export with minimal IAM), and Azure (multi-tenant app registration with admin consent flow – generates a service principal in the client tenant, used in an ARM template to provision necessary resources for cost data access)
- Built multi-cloud account validation service – AWS checks, GCP BigQuery, and Azure (RBAC, Cost Management, Blob) – deployed with Docker and Terraform
- Developed real-time alerting module with OpenSearch monitors, configurable thresholds, and webhook notification channels
- Implemented GCP pre-onboarding validator verifying BigQuery billing export access across projects; deployed on Kubernetes with Helm chart
- Built proof-of-concept AI cost estimation API using OpenAI GPT-4o-mini with structured JSON output and a multi-step pipeline (QueryParser → ParameterValidator → PromptEnhancer → OutputValidator)
- Deployed microservices on **AWS ECS** with RDS MySQL, SQS, and OpenSearch integration; authored provisioning scripts and Ansible playbooks

Stack: Python, FastAPI, SQLAlchemy, AWS (ECS, RDS, Cognito, Lambda, SQS, Athena, OpenSearch), GCP (BigQuery, IAM, GKE), Azure (Cost Management, ARM Templates), MySQL, Docker, Terraform, Kubernetes, Helm, OpenAI

Client Infrastructure Monitoring (ABAD, Poomkudy, Muralaya)

2024 – Present

- Managing end-to-end observability for multiple client environments: **Prometheus** alerting, **Grafana** dashboards, **Loki** log aggregation, and **Beyla** eBPF auto-instrumentation (zero-code overhead)
- Configured **Nginx** reverse proxy, SSL termination, and load balancing; maintained Node Exporter host-level metrics (CPU, memory, disk, network)

Stack: Prometheus, Grafana, Loki, Node Exporter, Beyla (eBPF), Nginx, Linux

Full-Stack Developer

Community Development Authority, Dubai – Official Government Project

Dec 2025
Remote

- Built and deployed the official CDA platform (dcsl.cda.gov.ae) that earned a **Guinness World Record** for highest online member participation
- Developed and **Dockerised** Next.js 15 app with multi-stage builds for Azure; implemented full i18n (English/Arabic RTL), password-less magic-link auth, and PostgreSQL schema with audit logging

Stack: Next.js 15, TypeScript, PostgreSQL, Docker, Azure, SendGrid, next-intl, reCAPTCHA

Contract Full-Stack Developer

Exotic-Green – Austria

2024
Remote • Contract • Sole Developer

- Sole developer for a full-stack warehouse management platform built for an Austrian supermarket chain
- Built **FastAPI** backend with inventory management, batch/best-before-expiry tracking, compliance protocols, and APScheduler-based auto-logger (cron scheduling, exponential backoff retry, Austrian timezone, graceful degradation)
- Developed **Next.js 14** frontend with webcam barcode scanning (zxing), PDF report generation, and AWS Cognito OAuth2 + email/password authentication with role-based route protection
- Led complete v2 redesign in **Next.js 16** – introduced Vitest unit tests, Playwright E2E tests, and web-vitals monitoring (LCP, FID,

- CLS, FCP, TTFB) from scratch
- Containerised full platform (FastAPI, Next.js, PostgreSQL, Redis) with Docker Compose for dev and production environments

Stack: Python, FastAPI, Next.js 14/16, TypeScript, PostgreSQL, Redis, AWS Cognito, shadcn/ui, Tailwind CSS, Docker

Full-Stack Developer
[Xmigrate](#)

Jun 2023 – Nov 2023
Remote

- Contributed to FastAPI backend covering org/project/blueprint management, target cloud configuration (AWS/Azure/GCP/OCI), agent registration, action lifecycle, and billing modules
- Integrated **Keycloak** for identity management: OAuth2/OIDC flows, SSO, custom themes, and token exchange; deployed on **AWS ECS Fargate** with RDS and S3
- Built Next.js 14 dashboard with React Flow topology diagrams and ApexCharts; developed SaaS frontend with Radix UI, TanStack Table, and PDF invoice generation via react-pdf
- Validated live VM migration workflows end-to-end across cloud environments, verifying agent communication and replication correctness

Stack: Python, FastAPI, Next.js 14, TypeScript, PostgreSQL, Keycloak, AWS (ECS Fargate, RDS, S3), Tailwind CSS, Cloudflare

Web Developer Intern
[Xmigrate](#)

May 2023 – Jun 2023
Remote

- Developed FastAPI endpoints for the cloud migration backend; contributed to API-frontend integration and data flow validation

EDUCATION

Bachelor of Computer Application (BCA)

2020 – 2023

Mahatma Gandhi University | Ernakulam, Kerala, India

TECHNICAL SKILLS

Backend	Python (FastAPI, Flask, Django), Go (Gin), Node.js, TypeScript, REST APIs, Microservices, WebSocket
Databases	PostgreSQL, MySQL, SQLAlchemy, MongoDB, OpenSearch, Redis, DynamoDB
Cloud & DevOps	AWS (ECS, EKS, RDS, Lambda, S3, Cognito, SQS, OpenSearch, Athena, Glue, ALB, VPC, IAM, ECR, Code-Pipeline), GCP (BigQuery, IAM, GKE), Azure (Cost Management, ARM Templates, Multi-tenant App Registration), Docker, Terraform, Kubernetes, Helm, Ansible, GitHub Actions
Frontend	Next.js (14/15/16), React.js, Tailwind CSS, shadcn/ui, Radix UI, MUI
Auth & Security	OAuth2, OIDC, JWT, AWS Cognito, Keycloak, Secretless Architecture, Azure AD Admin Consent, Multi-tenant RBAC
AI / ML	OpenAI (Structured JSON, MCP Pipeline), Amazon Bedrock, TensorFlow/Keras, Computer Vision (OpenCV); (LangChain, Stable Diffusion)
Tools	Git, Linux, Nginx, Shell Scripting, Vite, Docker Compose, (Jenkins)

COMMUNITY INVOLVEMENT

CCL (Continue Collaborate & Learn)	Mentor – internal community; mentored 50+ candidates across networking, Python, and cloud technologies. Trained engineering interns at Thoughtroutes on the same topics.
AWS Community Group, Kochi	Active member; volunteered at AWS Community Day Kochi 2025 (Dec 2025).
CNCF Community Chapter, Kochi	Active member; volunteered at Kubernetes Community Day (KCD) 2026 (Apr 2026). Regular attendee at community meetups across both groups.

blitz	Personal project (Rust) – HTTP & WebSocket stress testing CLI. Built with tokio, reqwest, and tokio-tungstenite; features real-time terminal dashboard, configurable RPM, latency percentiles (p50/p95/p99), multi-URL YAML pipelines, and JSON reports. Actively adding per-worker proxy rotation and Tor/onion routing. MIT licensed.
Xmigrate	Open source cross-cloud VM migration platform. Contributed FastAPI backend features and validated xm-replicator (Go, eBPF block-level replication) end-to-end across cloud environments.
SocialSim	(In development) Multi-agent social simulation: multi-source identity discovery (Bluesky, HN, Reddit), dynamic knowledge graph with scenario ontology, CAMEL-AI OASIS integration, and real-time WebSocket monitoring. Exploring LLM-driven agent orchestration as an SRE reliability analogue.

KEY PROJECTS

Multi-Cloud Cost Management Platform Python, FastAPI, React 18, AWS (ECS, RDS, SQS, Lambda, OpenSearch), GCP (BigQuery, IAM), Azure (Cost Management, ARM), MySQL, Terraform, Docker, OpenAI Production enterprise SaaS for cost governance across AWS, Azure, and GCP. Owned the entire backend: multi-tenant API server with per-tenant schema isolation, a 20+ pipeline ingestion engine (SQS-orchestrated), secretless multi-cloud onboarding (IAM roles on AWS, service accounts on GCP, Azure AD multi-tenant app registration with admin consent + ARM template provisioning), multi-cloud validation (Lambda/Terraform), OpenSearch alerting, and an AI cost estimation proof-of-concept using GPT-4o-mini.	Thoughtroutes · CloudPositive
Dubai Communicates in Sign Language (DCSL) Next.js 15, TypeScript, PostgreSQL 16, Tailwind CSS, next-intl, SendGrid, reCAPTCHA v2, Microsoft Azure, Docker Official government platform (dcsl.cda.gov.ae) that earned a Guinness World Record for highest online member participation. Bilingual (English/Arabic RTL), passwordless auth, event/quiz management, PostgreSQL schema with audit logging, Dockerised on Azure.	Thoughtroutes · CDA, Dubai Government · Guinness World Record
Warehouse Management System Python, FastAPI, Next.js 16, TypeScript, PostgreSQL, Redis, AWS Cognito, Docker Full-stack warehouse platform for an Austrian supermarket chain. Dockerised deployment, APScheduler compliance logging, barcode scanning, PDF reports, and full v2 rebuild with automated testing and web-vitals monitoring.	Thoughtroutes · Exotic Green
Cross-Cloud VM Migration Platform Python, FastAPI, Next.js 14, TypeScript, Keycloak, PostgreSQL, AWS ECS Fargate, Cloudflare Live VM migration SaaS with eBPF-based block-level replication. Contributed to FastAPI backend (27+ modules), Keycloak SSO, Next.js topology dashboards, and SaaS frontend on Cloudflare.	Xmigrate
AI-Powered Medicine Reminder Bot Python, AWS Durable Functions, Amazon Bedrock, AWS Lambda, Telegram Bot API Serverless Telegram bot using AWS Durable Functions and Amazon Bedrock. Stateless/stateful two-Lambda architecture with checkpoint/replay for execution state recovery and structured multimodal prompt validation – zero idle cost.	Personal Project

LANGUAGES

English (Fluent) | Malayalam (Native) | Hindi (Conversational)